

Transformation Evidence for Congruence and Similarity

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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| 1. the length of $AB = 5$, the length of the image $A'B' = 5$ | 7. the length of $GH = 5$, the length of $G'H' = 15$, the scale factor $= 3$ |
| 2. the length of $CD = 10$, the length of the image $C'D' = 10$ | 8. the slope of $JK = \frac{3}{4}$, the slope of the image $J'K' = \frac{3}{4}$ |
| 3. the ratio of the shortest sides $= 3$, the ratio of the middle sides $= 3$, the ratio of the longest sides $= 3$ | 9. the length of the base of the first triangle $= 3$, the length of the base of the image $= 3$, — |
| 4. the slope of $EF = 1 \cdot \frac{1}{2}$, the slope of the image $E'F' = 1 \cdot \frac{1}{2}$ | 10. the longest side of the small triangle $= 13$, the longest side of the large triangle $= 39$, the scale factor $= 3$ |
| 5. the length of the base of $P = 4$, the length of the base of $Q = 4$, — | 11. the slope of $BC = -\frac{3}{4}$, the slope of the image $B'C' = -\frac{3}{4}$, — |
| 6. the base of the first triangle $= 6$, the base of the second triangle $= 3$, — | 12. the length of $AB = 4$, the length of $DE = 6$, the ratio DE to $AB = \frac{3}{2}$, — |
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What is verified

7 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 5, 6, 9, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.G.A – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 33 tagged checks.

1. $A(1, 1)$, $B(4, 5)$ translated 5 right to $A'(6, 1)$, $B'(9, 5)$. Find AB and $A'B'$.

Solution: $AB = \sqrt{3^2 + 4^2} = \sqrt{25}$ and $A'B' = \sqrt{3^2 + 4^2} = \sqrt{25}$. A translation shifts both endpoints the same way, so the rise and the run are unchanged.

$AB = 5$	$A'B' = 5$
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2. $C(1, 2)$, $D(7, 10)$ reflected in the y -axis to $C'(-1, 2)$, $D'(-7, 10)$. Find CD and $C'D'$.

Solution: $CD = \sqrt{6^2 + 8^2} = \sqrt{100}$. For the image the run is $-7 - (-1) = -6$, and squaring removes the sign, so $C'D' = \sqrt{36 + 64} = \sqrt{100}$.

$$\boxed{CD = 10 \quad C'D' = 10}$$

3. Triangles with sides 3, 4, 5 and 9, 12, 15: divide each larger side by its match.

Solution: $\frac{9}{3} = 3$, $\frac{12}{4} = 3$, $\frac{15}{5} = 3$. All three ratios agree, which is exactly the evidence that the triangles are similar with scale factor 3.

$$\boxed{3 \quad 3 \quad 3}$$

4. $E(2, 5)$, $F(6, 7)$ rotated 180° about the origin to $E'(-2, -5)$, $F'(-6, -7)$. Compare the slopes.

Solution: Slope of $EF = \frac{7-5}{6-2} = \frac{2}{4}$. Slope of $E'F' = \frac{-7-(-5)}{-6-(-2)} = \frac{-2}{-4}$. Both signs cancel, giving the same value.

$$\boxed{\text{slope } EF = \frac{1}{2} \quad \text{slope } E'F' = \frac{1}{2}}$$

5. Triangle P : $(1, 2)$, $(5, 2)$, $(1, 5)$. Triangle Q : $(1, -2)$, $(5, -2)$, $(1, -5)$. Find both bases and name the transformation.

Solution: Both bases are horizontal, so each length is the gap between the two x -values: $5 - 1 = 4$ in each triangle.

$$\boxed{\text{base of } P = 4 \quad \text{base of } Q = 4}$$

Transformation (instructor-judged). Every vertex keeps its x -coordinate and has its y -coordinate negated, so the map is a reflection in the x -axis. A correct response also says why that settles congruence: a reflection is a rigid motion, so it preserves every length and angle, and the two triangles must be congruent. Naming the reflection without the length argument is only half the answer.

6. Is $(0, 0)$, $(6, 0)$, $(0, 4)$ congruent to $(0, 0)$, $(3, 0)$, $(0, 2)$?

Solution: Both bases lie on the x -axis, so read them straight off: the first is 6 and the second is 3.

$$\boxed{\text{first base} = 6 \quad \text{second base} = 3}$$

Judgement (instructor-judged). A correct response says the student is wrong. Congruent figures need equal matching lengths, and $6 \neq 3$. The right description is *similar*: every side of the second triangle is half the matching side of the first, so the two are the same shape at half the size. Accept any wording that rejects congruence on the unequal lengths and names the halving.

7. $G(0,0)$, $H(4,3)$ dilated about the origin to $G'(0,0)$, $H'(12,9)$. Find both lengths and the scale factor.

Solution: $GH = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$ and $G'H' = \sqrt{12^2 + 9^2} = \sqrt{225} = 15$. The scale factor is the image length over the original: $\frac{15}{5}$.

$$\boxed{GH = 5 \quad G'H' = 15 \quad k = 3}$$

8. $J(2,1)$, $K(6,4)$ dilated by 2 to $J'(4,2)$, $K'(12,8)$. Compare the slopes.

Solution: Slope of $JK = \frac{4-1}{6-2} = \frac{3}{4}$. Slope of $J'K' = \frac{8-2}{12-4} = \frac{6}{8}$, the same fraction. The dilation doubled both the rise and the run, so the ratio could not change.

$$\boxed{\text{slope } JK = \frac{3}{4} \quad \text{slope } J'K' = \frac{3}{4}}$$

9. $(2,1)$, $(5,1)$, $(2,3)$ maps to $(-2,1)$, $(-5,1)$, $(-2,3)$. Find both bases and name the transformation.

Solution: The first base runs from $x = 2$ to $x = 5$, a length of 3; the image base runs from $x = -2$ to $x = -5$, also a length of 3.

$$\boxed{\text{first base} = 3 \quad \text{image base} = 3}$$

Transformation (instructor-judged). Each x -coordinate is negated and each y -coordinate is unchanged, so this is a reflection in the y -axis. A correct response adds that a reflection preserves distance, so matching sides stay equal and the triangles are congruent. A name alone is half the answer.

10. Right triangles from the origin to $(5,12)$ and to $(15,36)$. Find both longest sides and their ratio.

Solution: $\sqrt{5^2 + 12^2} = \sqrt{169} = 13$ and $\sqrt{15^2 + 36^2} = \sqrt{1521} = 39$. Dividing gives $\frac{39}{13}$, and the legs scale the same way ($15 = 3 \cdot 5$, $36 = 3 \cdot 12$), so the two triangles are similar.

$$\boxed{13 \quad 39 \quad \text{ratio} = 3}$$

11. $A(0,0)$, $B(4,0)$, $C(0,3)$ dilated by 2 to $A'(0,0)$, $B'(8,0)$, $C'(0,6)$. Compare the slopes of BC and $B'C'$.

Solution: Slope of $BC = \frac{3-0}{0-4} = -\frac{3}{4}$ and slope of $B'C' = \frac{6-0}{0-8} = -\frac{6}{8}$, which reduces to the same number.

$$\boxed{\text{slope } BC = -\frac{3}{4} \quad \text{slope } B'C' = -\frac{3}{4}}$$

Reason (instructor-judged). A correct response says the equal slopes make BC and $B'C'$ parallel, and that the same is true of every pair of matching sides, so the matching angles of the two triangles are equal. That is what makes the image similar to the original rather than a different shape.

12. ABC : $A(-4, -2)$, $B(0, -2)$, $C(-4, 1)$. DEF : $D(0, 0)$, $E(6, 0)$, $F(0, 4.5)$. Congruent, similar, or neither?

Solution: AB is horizontal from $x = -4$ to $x = 0$, so $AB = 4$; DE is horizontal from $x = 0$ to $x = 6$, so $DE = 6$. The ratio is $\frac{6}{4}$, which reduces to $\frac{3}{2}$. The other pair matches it too: $AC = 3$ and $DF = 4.5$, and $4.5 \div 3 = 1.5$ as well.

$AB = 4$	$DE = 6$	ratio = $\frac{3}{2}$
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Conclusion (instructor-judged). A correct response chooses *similar but not congruent* and supports it with both facts: the matching lengths are unequal (4 against 6), which rules out congruence, and every pair of matching sides is in the same ratio 3 : 2, which is what similarity requires. A choice with no computed support earns no credit.